

DIHS Tephra Correlator - Getting Started Guide

<https://tephracorrelator.geo3bcn.csic.es/> is the user-friendly implementation of the v1.0.0 version of the DIHS_Correlator Python package (Aymerich et al., 2026), openly developed at https://github.com/jaymerichni/DIHS_Correlator

If you detect any bugs or have any difficulties running the DIHS Tephra Correlator, please feel free to contact jaymerichni@gmail.com

This quick tutorial is meant to show how to use the main tools of the package in a straightforward manner. Keep in mind that you can also install the programme and navigate this browser-based interface from your machine, allowing you to allocate as many resources to the analysis as you desire (the currently deployed online version limits the usage to avoid over-saturation of the CSIC servers). Find more information about the package installation in Aymerich et al. (2026).

1. Upload Dataset



DIHS TEPHRA ANALYSIS WORKSPACE

DIHS Tephra Correlator

POWERED BY DIHS_CORRELATOR V1.0.0

1. Upload Dataset

CSV dataset

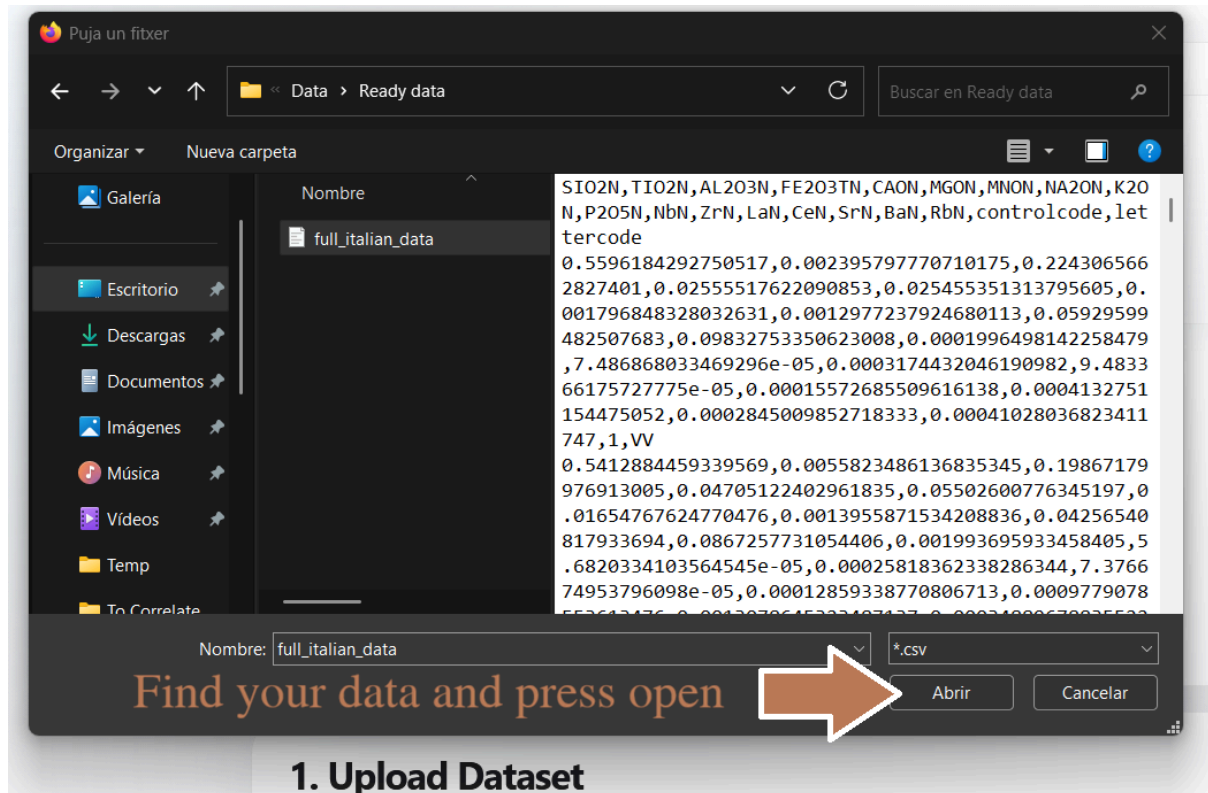
 **Press this button to browse files from your machine**

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When entering the webpage for the first time you will notice that you can only do one thing: upload data in table format (for now, only .csv files are supported). Click the highlighted button and navigate your files.



Once you have found the right .csv file for your correlation, click *Open* and then *Load CSV*.



Afterwards, you will get two new modules (**2. Dataset Preview** and **3. Configure Analysis**).

2. Dataset Preview

The first one is simply a preview of the table you uploaded.

2. Dataset Preview

full_italian_data.csv | 3,907 rows x 19 columns

MNON	NA2ON	K2ON	P2O5N	NBN	ZRN	LAN	CEN	SRN	BAN	RBN	CONTROLCODE	LETTERCODE
0.001298	0.059296	0.098328	0.000200	0.000075	0.000317	0.000095	0.000156	0.000413	0.000285	0.000410	1	VV
0.001396	0.042565	0.086726	0.001994	0.000057	0.000258	0.000074	0.000129	0.000978	0.001308	0.000349	1	VV
0.001298	0.063391	0.097332	0.000200	0.000070	0.000334	0.000111	0.000179	0.000370	0.000156	0.000402	1	VV
0.001298	0.061088	0.098320	0.000299	0.000070	0.000334	0.000107	0.000176	0.000422	0.000220	0.000398	1	VV
0.001298	0.060586	0.098115	0.000299	0.000068	0.000328	0.000106	0.000174	0.000449	0.000266	0.000394	1	VV
0.001297	0.056581	0.096697	0.000599	0.000065	0.000316	0.000102	0.000166	0.000543	0.000423	0.000384	1	VV
0.001298	0.057396	0.096126	0.000499	0.000065	0.000312	0.000102	0.000154	0.000432	0.000257	0.000387	1	VV
0.001297	0.053878	0.094686	0.000898	0.000064	0.000297	0.000099	0.000163	0.000653	0.000603	0.000374	1	VV
0.001297	0.049085	0.092184	0.001097	0.000059	0.000284	0.000096	0.000159	0.000680	0.000687	0.000368	1	VV
0.001396	0.044764	0.090127	0.001795	0.000055	0.000265	0.000088	0.000151	0.000837	0.000964	0.000359	1	VV

Make sure that the table contains the columns you need for the correlation (mainly each geochemical feature and the class column). If you detect any other column that is not required you can exclude it from the analysis later on without changing the original file.

Data Requirements

The uploaded data must have at least a numerical column for each one of the geochemical features you wish to use for the correlation (e.g. SiO₂, Zr, Sr, La/Sc, etc.) and another string column that relates each row to the class it belongs to. The name of the class column does not matter as long as it refers to the specific geological entities you wish to compare your uncorrelated data to (e.g. unit, member, volcano, volcanic province, region, etc.). The sample of unknown origin you wish to compare to its potential sources, must be concatenated in the table as its own class with a concrete name you will specify later in the **3. Configure Analysis** module.

3. Configure Analysis

This module allows the configuration of the input parameters that control the analyses. For the example data being used in this guide, the next figure shows what the module looks like right after loading. At first, the parameters are not well chosen for the desired analysis and need toggling.

3. Configure Analysis

Analysis workflow: Single-model DIHS run
Model: agglomerative
Class column: controlcode
Unknown sample/class: 0
Transform: clr

Max depth: 100
Random state: 0
☐ Use random state
☒ Compute pairwise matrices
☐ Create plots

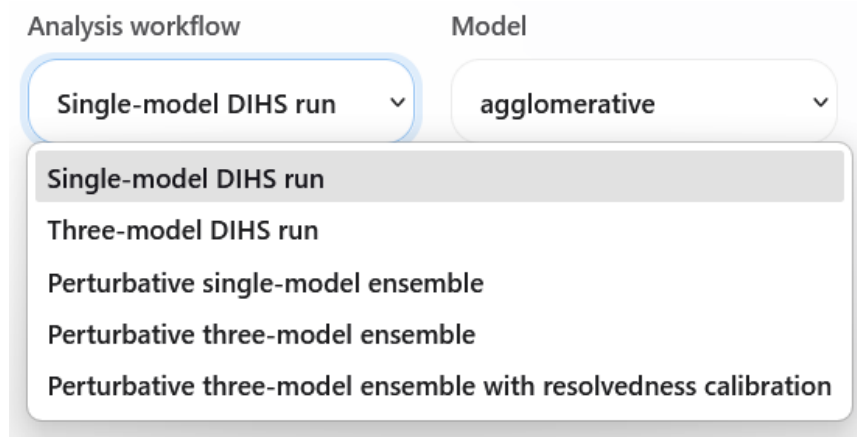
Exclude columns from numeric features:
SIO2N
TIO2N
AL2O3N
FE2O3TN
CAON

Run Analysis

Each *Analysis workflow* enables different parameters that configure the runs.

<i>Analysis workflow (dropdown menu)</i>	
Options	Description
<i>Single-model DIHS run</i>	One model to choose from the <i>Model</i> dropdown menu drives a single run without analytical uncertainty propagation.
<i>Three-model DIHS run</i>	All three available models drive each a single run without analytical uncertainty propagation.
<i>Perturbative single-model ensemble</i>	One model to choose from the <i>Model</i> dropdown menu drives as many Monte Carlo iterations as specified in the <i>Perturbative iterations</i> field, propagating the specified analytical uncertainty/-ies in the <i>Major element error</i> or/and <i>Trace element error</i> fields.
<i>Perturbative three-model</i>	All three available models drive the same amount of

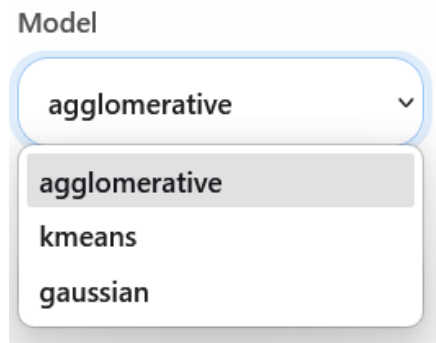
<i>ensemble</i>	Monte Carlo iterations specified in the <i>Perturbative iterations</i> field, propagating the specified analytical uncertainty/-ies in the <i>Major element error</i> or/and <i>Trace element error</i> fields.
<i>Perturbative three-model ensemble with resolvedness calibration</i>	All three available models drive the same amount of Monte Carlo iterations specified in the <i>Perturbative iterations</i> field, propagating the specified analytical uncertainty/-ies in the <i>Major element error</i> or/and <i>Trace element error</i> fields. Afterwards, the desired amount of <i>Pseudo-unknown iterations</i> calibrate the resolvedness of the association.



The *Model* dropdown menu allows you to choose which clustering model will drive recursive binary clustering for the *Single-model DIHS run* and the *Perturbative single-model ensemble* workflows.

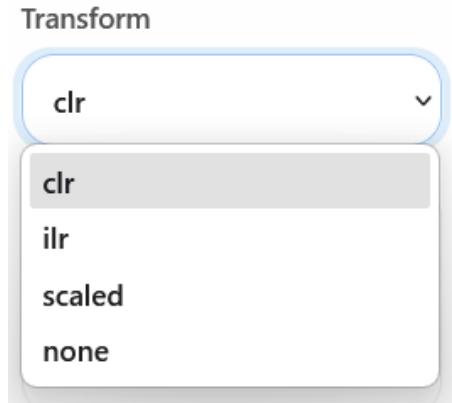
<i>Model</i> (dropdown menu)	
Options	Description
<i>agglomerative</i>	Recursive binary Agglomerative-Ward clustering drives the creation of the tree (or trees depending on the selected <i>Analysis workflow</i>) from which the correlation metrics derive.
<i>kmeans</i>	Recursive binary K-Means clustering drives the creation of the tree (or trees depending on the selected <i>Analysis workflow</i>) from which the correlation metrics derive. If selected, the field <i>Random state</i> determines the initialization seed for the clustering model.

<i>gaussian</i>	Recursive binary Gaussian Mixture Model clustering drives the creation of the tree (or trees depending on the selected <i>Analysis workflow</i>) from which the correlation metrics derive. If selected, the field <i>Random state</i> determines the initialization seed for the clustering model.
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Different transformations, including CoDa log-ratios (Aitchison, 1982) and Z-scaling are implemented and are available for each workflow analysis.

<i>Transform (dropdown menu)</i>	
Options	Description
<i>clr</i>	Implements the Centered-Log-Ratio transformation on the selected numerical features of the data. Use only with compositional data that closes near a constant sum.
<i>ilr</i>	Implements the Isometric-Log-Ratio transformation on the selected numerical features of the data. Use only with compositional data that closes near a constant sum.
<i>scaled</i>	Implements Z-scaling on the selected numerical features of the data (mean of 0 and 1 standard deviation).
<i>none</i>	No transformation or scaling is applied to the data.



Beyond these three dropdown menus, there exist eight other toggles/fields common to all analysis workflows:

Basic parameters		
Parameter	Type	Description
<i>Class column</i>	Dropdown menu	Choose the column name that encodes class identity for each row in your table. The menu will show all the columns it detects, the same ones you see in the table preview.
<i>Unknown sample/class</i>	Dropdown menu	Choose the name you assigned to the sample of unknown origin you want to correlate. If you did not choose the right <i>Class column</i> value before, you will not see it on the list.
<i>Max depth</i>	Field (integer)	This parameter controls how deep the recursive clustering can go.
<i>Random state</i>	Field (integer)	This parameter only works when <i>Use random state</i> is on. It represents the initialization seed of the K-Means and Gaussian Mixture clustering models and therefore does nothing for single-model configurations with Agglomerative-Ward, which is deterministic.
<i>Use random state</i>	Checkbox	Activates/deactivates the use of the initialization seed specified in <i>Random state</i> for the K-Means and Gaussian

		Mixture models.
<i>Compute pairwise matrices</i>	Checkbox	If checked on, HS and DIHS are not only computed between the unknown class and the potential sources but also between them. The stopping conditions, however, are controlled by the unknown class.
<i>Create plots</i>	Checkbox	If checked on, all the plots relevant to the selected workflow will be generated and shown in the results module.
<i>Exclude columns from numeric features</i>	Multiple-option dropdown menu	Choose all the columns you wish to leave out from the feature space of your analysis. If there are any numerical columns left by mistake they will most likely affect the results without raising an error.

The screenshot displays a software interface with several configuration sections:

- Analysis workflow:** A dropdown menu set to "Perturbative three-model e".
- Class column:** A dropdown menu set to "lettercode".
- Unknown sample/class:** A dropdown menu set to "Caio".
- Transform:** A dropdown menu set to "clr".
- Max depth:** A numeric input field set to "100".
- Random state:** A numeric input field set to "0", with a note: "Only applied when the checkbox 'Use random state' is enabled."
- Use random state:** An unchecked checkbox.
- Compute pairwise matrices:** A checked checkbox.
- Create plots:** A checked checkbox.
- Exclude columns from numeric features:** A list box containing "SrN", "BaN", "RbN", "controlcode" (highlighted), and "lettercode".

For the analysis workflows that involve analytical uncertainty propagation through perturbative resampling, there exist seven other parameters:

Perturbative workflow parameters		
Parameter	Type	Description
<i>Perturbative iterations</i>	Field (integer)	Choose the number of ensemble runs you wish your perturbative analysis to have. Keep in mind that large numbers will result in longer computing times.
<i>Major element error</i>	Field (float)	Type the analytical uncertainty envelope* you wish to propagate across the ensemble for the major element group (or any other group of

		features you organize under that name, selecting them in the <i>Major columns</i> dropdown menu).
<i>Trace element error</i>	Field (float)	Type the analytical uncertainty envelope* you wish to propagate across the ensemble for the trace element group (or any other group of features you organize under that name, selecting them in the <i>Trace columns</i> dropdown menu).
<i>Perturbation seed</i>	Field (integer)	Control the seed of the random sampling of values within the analytical uncertainty envelope. Serves for reproducibility purposes.
<i>Integration depth</i>	Field (integer)	Choose the depth at which you wish to compute the DIHS distribution of the ensemble. If no value is chosen the maximum common depth among all trees is used.
<i>Major columns</i>	Multiple-option dropdown menu	Select the columns that correspond to major elements in your feature space or any other group you wish to perturb within the <i>Major element error</i> envelope.
<i>Trace columns</i>	Multiple-option dropdown menu	Select the columns that correspond to trace elements in your feature space or any other group you wish to perturbate within the <i>Trace element error</i> envelope.

*0.02 or 0.1 values –for instance– represent 2% and 10% uncertainty envelopes, respectively.

Perturbative Ensemble Settings

Perturbative iterations

100

Major element error

0.02

Trace element error

0.1

Perturbation seed

Integration depth

Major columns

CAON

MGON

MNON

NA2ON

K2ON

P2O5N

Trace columns

K2ON

P2O5N

NbN

ZrN

LaN

CaN

Finally, the *Perturbative three-model ensemble with resolvedness calibration* analysis workflow requires configuration of the five parameters that control the final calibration step.

Calibration step parameters		
Parameter	Type	Description
<i>Pseudo-unknown iterations</i>	Field (integer)	Choose the number of sampling iterations you wish your calibration step to be based on.
<i>Pseudo-unknown sample size</i>	Field (integer)	Choose the number of rows/observations you wish to sample at each iteration for the pseudo-unknown tests.
<i>Pseudo-unknown random state</i>	Field (integer)	Control the seed of the random sampling of rows for the pseudo-unknown tests. Serves for reproducibility purposes.
<i>Target precisions</i>	Field (float)	Type any specific precision (resolvedness value) in decimal format to get the $\Delta DIHS$ threshold that would empirically satisfy that precision.
<i>Minimum runs above threshold</i>	Field (integer)	Choose the minimum number of pseudo-unknown runs above the desired threshold you wish the calibration step to take into account.

Resolvedness Calibration Settings

Pseudo-unknown iterations

Pseudo-unknown sample size

Pseudo-unknown random state

Target precisions

Minimum runs above threshold

100

17

1

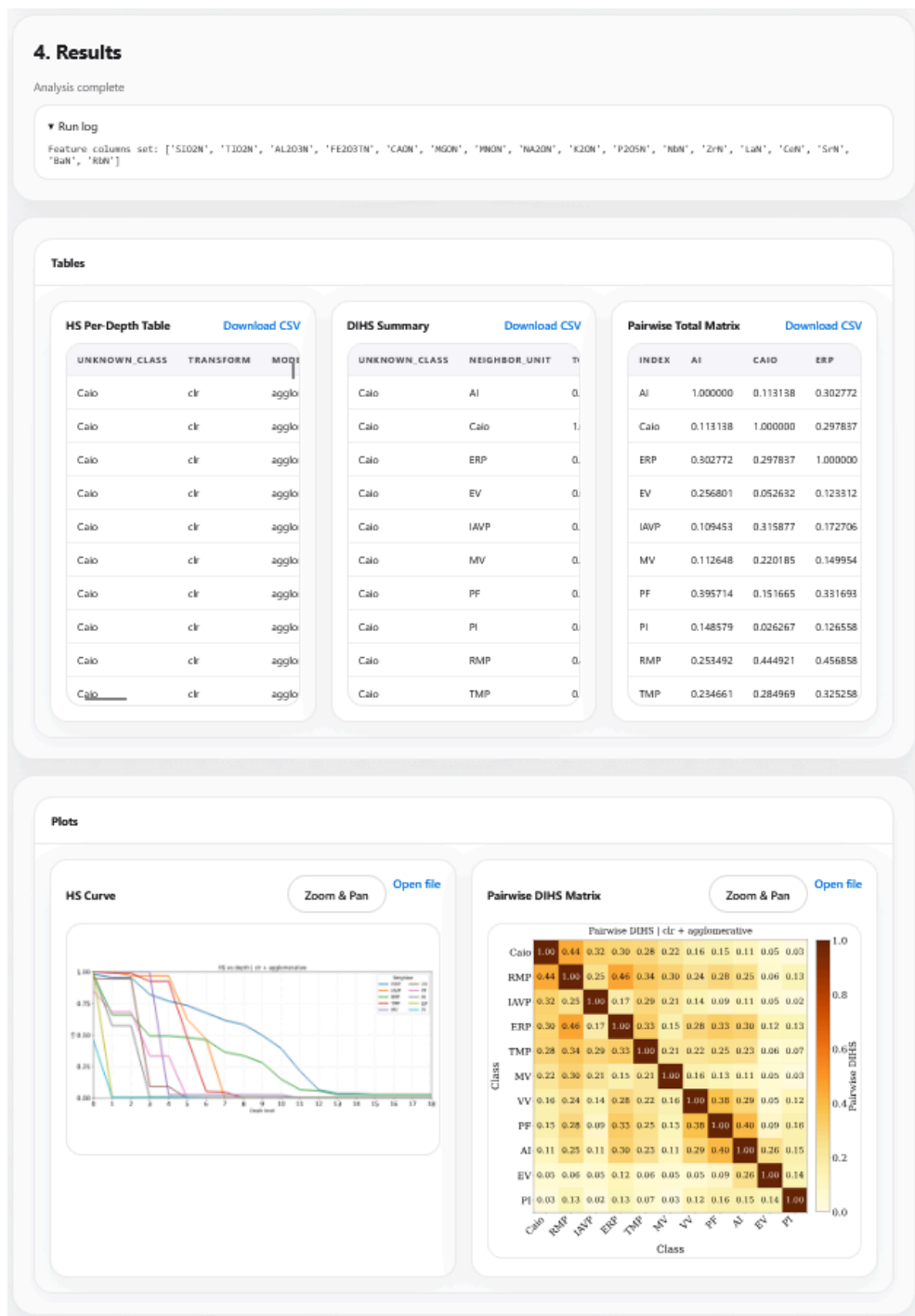
Comma-separated values, for example 0.95,0.9,0.85

Run Analysis

Click here to begin the analysis

4. Results

Once the analysis has been completed (the development can be monitored through a progress bar), new modules will appear with result tables (downloadable in .csv format) and plots (downloadable in .png format), if *Create plots* is enabled. The run log can also be accessed from the main module.



How to interpret the results

HS describes the association between the unknown and each candidate class at an individual level of the clustering hierarchy. DIHS integrates HS across hierarchical depth, so higher DIHS values indicate a stronger and more persistent association. Compare candidates by their DIHS values and inspect the HS curve to determine whether the association is sustained rather than restricted to a single depth.

Δ DIHS is the difference between the highest and second-highest DIHS values. A larger margin indicates clearer separation between the leading candidates, whereas a small margin indicates greater ambiguity. In the calibrated workflow, pseudo-unknown experiments estimate the Δ DIHS threshold associated with each requested precision or resolvedness level. These outputs rank associations within the supplied candidate dataset; the highest-scoring candidate is not, by itself, proof of geological correlation.

References

Aymerich J, Petrelli M, Ágreda-López M, et al (2026) DIHS_Correlator. Zenodo, <https://doi.org/10.5281/zenodo.21805857>, version 1.0.0. Computer software

Aitchison J (1982) The statistical analysis of compositional data. *Journal of the Royal Statistical Society: Series B (Methodological)* 44(2):139–160. <https://doi.org/10.1111/j.2517-6161.1982.tb01195.x>